



Digestomatic



1.1 Overview & Features

Digestomatic is a smart biodigester prototype that integrates IoT, AI, and automation to enhance user experience and the production of renewable gas. Using an ESP32, specialized gas sensors, temperature sensors and a locally hosted web dashboard, the system continuously monitors conditions inside the biodigester. A local LLM analyzes the data and answers questions regarding the digester and its performance. Furthermore, by utilizing a DC motor powered stirrer, the substrate is automatically mixed to maximize methane production.



Features:

- Safe and real-time CH₄ and CO₂ concentration and temperature monitoring;
- Digestomatic dashboard displaying digester metrics;
- Foria A.I for interpretations regarding performance and answering questions;
- Automatic stirrer mechanism;
- Optimized methane production;
- Downloadable measurement logs for further analysis;
- A more user friendly experience;
- Modular design.

What problems does Digestomatic solve?

A biodigester is a unique solution to two very big problems: (1) organic material being dumped and/or burned and (2) most gas production still being reliant on non-renewable systems. It tackles both very elegantly by utilizing [anaerobic digestion](#). However many biodigesters are passive systems, meaning they lack control systems. Furthermore, there is no safe way to monitor the conditions inside the digester without opening the tank. Digestomatic aims to solve both problems and go beyond by integrating AI.

Why Artificial Intelligence?

Some knowledge about anaerobic digestion and biogas is required to operate and maintain a biodigester. Even then, the gas production can stagnate if certain conditions are not met properly, such as temperature or substrate composition. By integrating a local LLM that can read measurements and give the interpretations of them, any causes of disruption can be identified and the user experience can be greatly improved. Furthermore, the user can prompt the AI with questions regarding the digester and it can give insightful, informational responses back.

1.2 Demo

Please watch the [Digestomatic Demo](#) where the features & use cases are explained. Below is a flowchart showcasing how the system works.

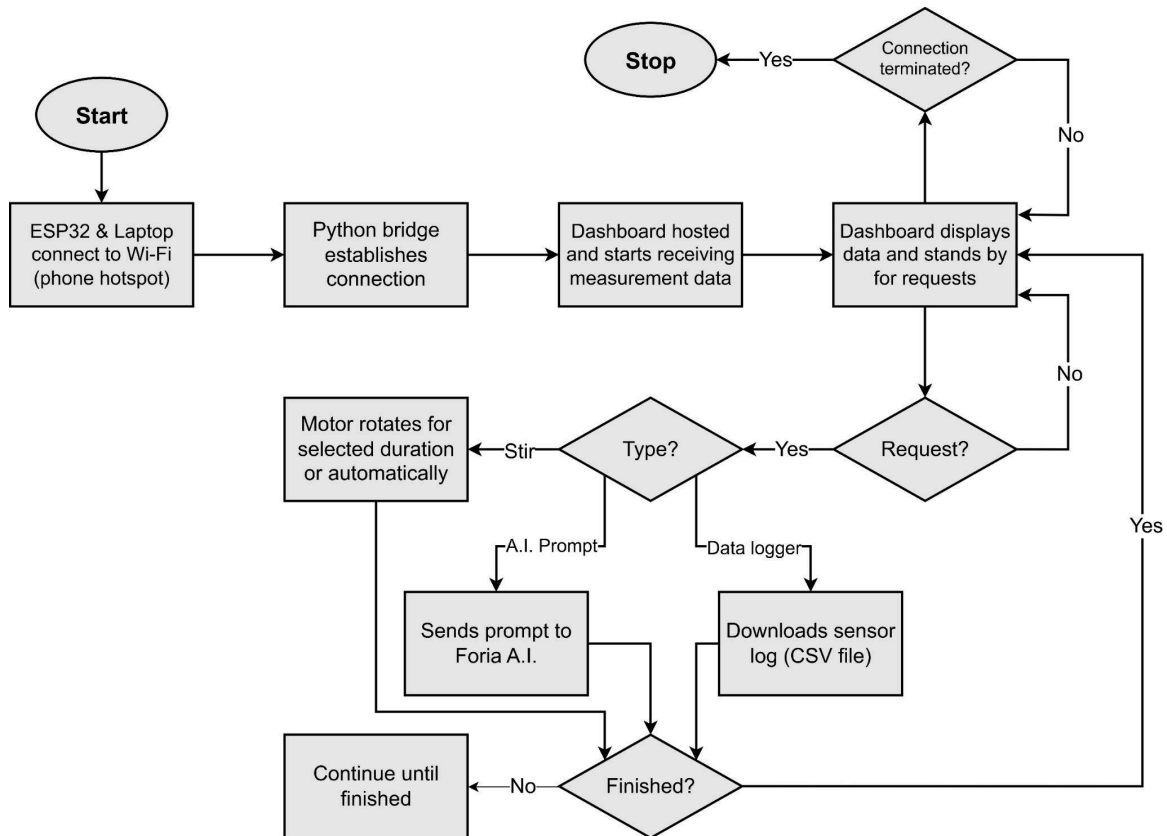


Figure: Flowchart showing how the logic of Digestomatic works.

1.3 Deliverables

Below is the list of deliverables for project Digestomatic:

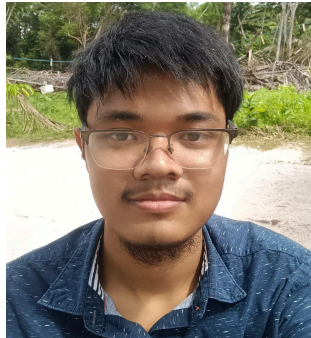
- Biodigester tank with inlet, control unit, gas outlet, and effluent outlet;
- Modular control unit with stirrer mechanism and monitor system;
- Back-end software for sensors, stirrer, data logging & A.I bridge;
- Web dashboard for displaying measurements and interfacing with Foria A.I. (Front-end);
- Foria: local LLM for questions and interpreting Digestomatic performance;
- Business model;
- Project poster;
- Project documentation including GitHub repo with all programs, assets and libraries;
- Pitch deck.

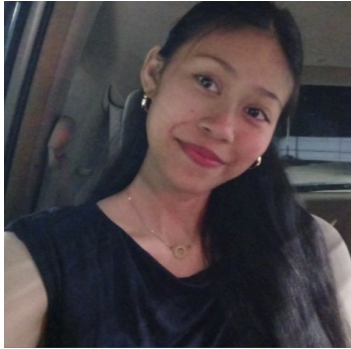
1.4 The Team & Contact

The team name is Eudaimonia ([εὐδαιμονία](#)), a Greek word which can be literally translated into *human flourishing* or the state or condition of *good spirit*. At its core, striving to do good for humanity. Team Eudaimonia consists of 5 AdeKUS students: Jozua Margaret, Riaasat Salarbux, Vaughn Sprangers, Christopher Tekam & Elijah Wongsonadi. 3 (Margaret, Salarbux & Sprangers) are enrolled in the Mechanical Engineering program and 2 (Tekam & Wongsonadi) in the Electrical Engineering Program. See the table below for more details.

Each of us are participating in the Hackomation because of interest in the challenge and to further develop our own technical skills. Participating allows us to delve into topics that are not part of our curriculum (or are introduced much later), such as Internet of Things, Artificial Intelligence, Robotics and Python.

Table: Team Eudaimonia Members and Information.

#	Personal Information		Roles	Photo
1	First Name	Jozua	Artist / Designer & Mechanical Engineer	
	Last Name	Margaret		
	Age	20		
	Residence	Noord		
	Email	jozuamargaret@gmail.com		
	Strengths	Blender & Mechanical Design		
2	First Name	Riaasat	Team Leader & All-Rounder	
	Last Name	Salarbux		
	Age	20		
	Residence	Westelijke Polders		
	Email	riaasatsalarbux80085@gmail.com		
	Strengths	Autodesk Fusion & Mechanical Design		
3	First Name	Vaughn	Front-end Developer & Mechanical Engineer	
	Last Name	Sprangers		
	Age	19		
	Residence	Flora		
	Email	sprangersv@gmail.com		
	Strengths	Design & Construction		

4	First Name	Christopher	Electrical Design Lead & Creative Director	
	Last Name	Tekam		
	Age	20		
	Residence	Van Pettenpolder		
	Email	christekam14@gmail.com		
	Strengths	Electrical Design & Modeling		
5	First Name	Elijah	Hardware & Back-end Technician	
	Last Name	Wongsonadi		
	Age	19		
	Residence	Van Pettenpolder		
	Email	e.wngs307@gmail.com		
	Strengths	Programming & Breadboarding		

2 Design & Model

2.1 Objectives

To design and develop a functional biodigester tank prototype that converts organic waste into biogas while ensuring safety and sustainability.

2.2 Steps Taken

- Design the physical structure of the biodigester.
- Develop a 3D model of the complete system.
- Select portable and affordable materials for the digester and supporting components. (A real biodigester would be too heavy and immovable)
- Design the inlet, outlet and biogas collection system and ensure the design minimizes gas leakage.
- Collect waste and build a second tank to cultivate bacteria.
- Integrate the automatic stirring mechanism into the biodigester.
- Design the enclosure for the electronic components.

2.3 Testing & Problems

The first iteration of the tank prototype was modeled and presented to Mr. Samuel Kuik, who has experience building biodigesters. After paying him a visit, he offered insightful feedback and instructions on how to improve the design. It required new holes to be drilled (for the inlet & gas outlet) and pipes to be fitted.

Drilling holes into the tank was difficult because everything had to be done at an angle and the thin walls lacked surface area to glue the pipes properly. In one instance, the hole for the outlet pipe was drilled too wide to be glued, requiring an improvised solution. The initial plan was to 3D print an angled uniseal to secure the pipe, but communication errors led to an undersized print. Under a tight deadline, the problem was solved by heating a PVC pipe into a flange with help from Assistant Workshop Manager Stefanito Soerotomo.

Achieving airtight seals at every pipe connection presented another challenge. While Polymax silicone sealant is typically applied around each connection, the enclosure housing the DC motor and electronics needed to remain removable while protecting the wiring from moisture. O-rings were initially purchased to allow the enclosure to be opened and closed securely while maintaining an airtight seal. However, the O-rings were too thick to fit properly. Due to time constraints, thin rubber salvaged from an old motorcycle wheel inner

tube was used instead, which, when applied with Vaseline, successfully created an airtight seal.

Connecting the DC motor to the paddle stirrer also presented a challenge. However, based on advice from Mr. H. Sariman (WB Machine Design lecturer), this was resolved by placing the motor shaft inside a hollow aluminum tube, which served as the base for the paddle stirrer, and securing it with a nut and bolt. The torque is transferred via friction between the tightened bolt and the shaft.

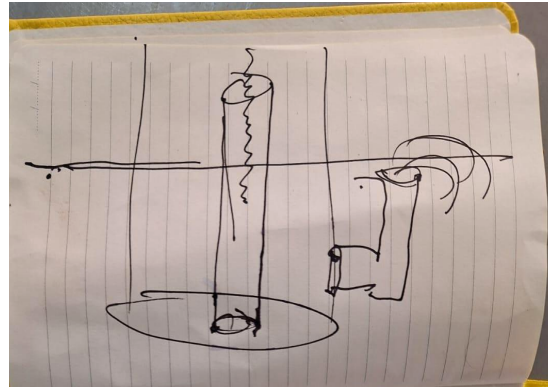
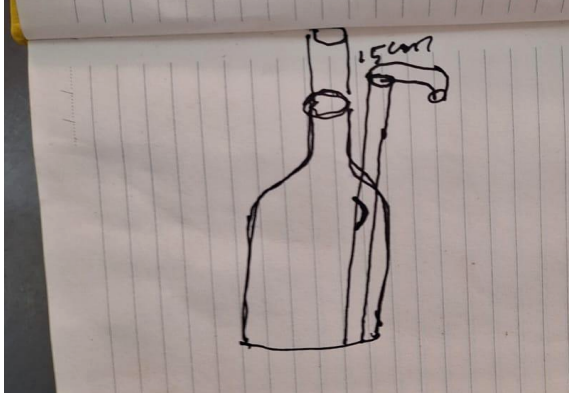
During the testing of the stirrer mechanism, several mechanical issues were encountered and resolved:

- **Stirrer Misalignment:** The stirrer was not perfectly aligned, causing instability during operation. This issue was resolved by carefully adjusting the motor to a specific position that minimized vibration and ensured smooth rotation.
- **Insufficient Motor Mount Strength:** The DC motor mount was initially unable to withstand the torque generated by the propeller. To reinforce the assembly, a substantial amount of epoxy adhesive was applied to securely bond the propeller and the connecting rod, improving the structural strength of the system.
- **Loose Propeller Coupling Screw:** The screw connecting the motor shaft to the propeller repeatedly loosened due to motor vibrations. The connection was reinforced to prevent the screw from loosening during operation.
- **DS18B20 Sensor Protection:** The DS18B20 temperature sensor was positioned close to the rotating propeller, creating a risk of physical contact. The sensor was secured using tape and superglue to keep it firmly in place and prevent interference with the propeller during operation.

During the electronics integration, the original power source was found to be insufficient because it could only provide 500 A at 12 V, while the DC motor required approximately 600 A, and the motor driver could not step the voltage down from 12 V to 5 V for the control electronics. This issue was resolved by replacing the power supply with a suitable PSU provided by Mr. Kartopawiro. In addition, the initial soldered wire connections insulated with electrical tape proved unreliable, as the tape frequently slipped off, causing occasional short circuits and overheating. These connections were replaced with screw terminal blocks to provide a more secure and reliable electrical connection.

Due to time constraints, there was not enough time to grow the methane bacteria culture after the main bioreactor finished. Therefore, a second tank was created, filled with cow manure, and spray-painted black to cultivate a good inoculant for later use.

2.4 Proof of Work



Figures: Sketches made by Mr. S. Kuik for the 2nd iteration of the main tank. (16/5/2026)



Figure: 2nd iteration of the main tank. (6/7/2026)



Figure: Our team working on the 2nd iteration of the main tank. (6/6/2026)



Figure: 3rd iteration of the main tank.
(31/7/2026)



Figure: 2nd tank being spray-painted
black. (6/7/2026)



Figure: 2nd tank containing methane bacteria
culture (13/7/2026)



Figure: 2nd tank after 12 days (25/7/2026)



Sideview
Figure: Final iteration of the
main tank
(5/8/2026)



Top angled view
Figure: Final iteration of the
main tank
(5/8/2026)

2.5 Files

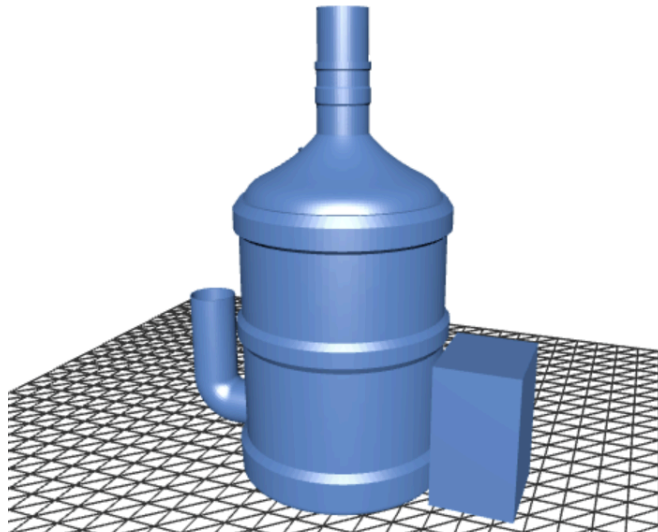


Figure: [First iteration digester tank model](#).

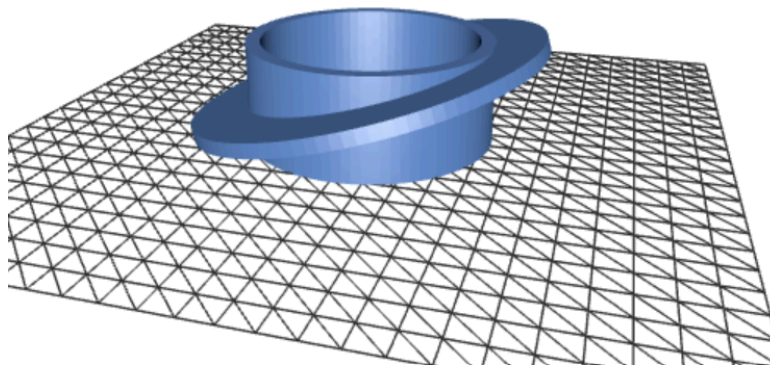


Figure: [Angled Uniseal](#) (ended up not being used).

2.6 Credits

Special thanks to:

- Mr. Samuel Kuik for his instructions on how to improve upon the first version of the biodigester.
- Mr. Stefanito Soerotomo for his assistance & Mr. H. Sariman for his advice.
- Mr. C. Kartopawiro for the PC PSU.

3 Embedded Solution

3.1 Objectives

To develop an embedded monitoring and stirrer system that continuously measures biodigester conditions, optimizes methane production and provides users with data through an IoT platform.

3.2 Steps Taken

- Collect ESP32, wires and other electronics.
- Wire and interface temperature sensors, DC motor and gas sensors with ESP32
- Identify necessary Arduino & Python libraries and download them.
- Develop the automatic stirring control algorithm.
- Develop logic for measuring temperature, CO₂ and CH₄ concentrations.
- Develop bridge backend for A.I.
- Set up communication backend between ESP32 and the dashboard.
- Develop data logging functionality.

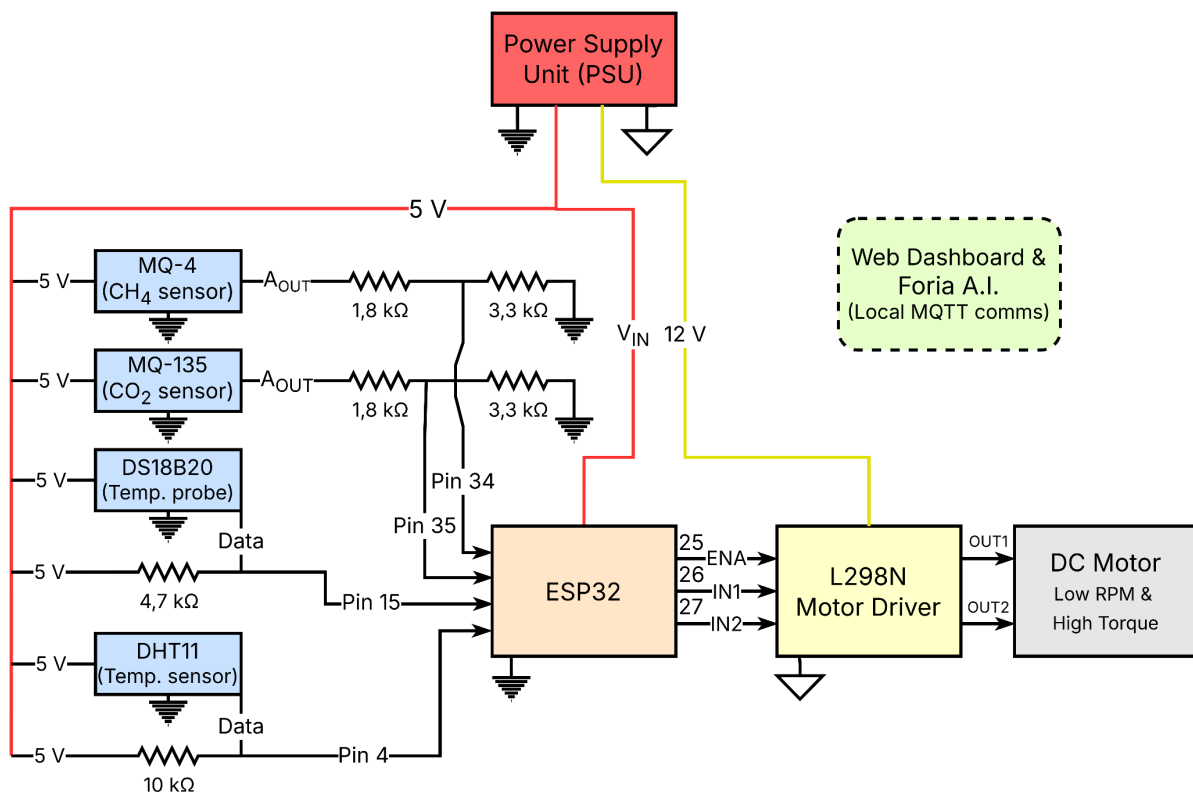


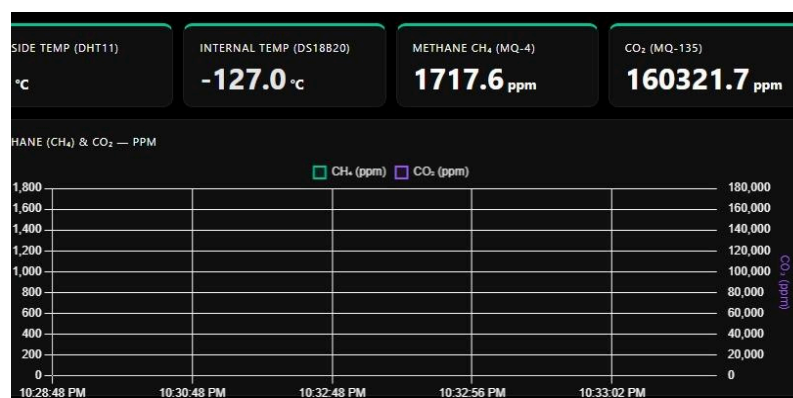
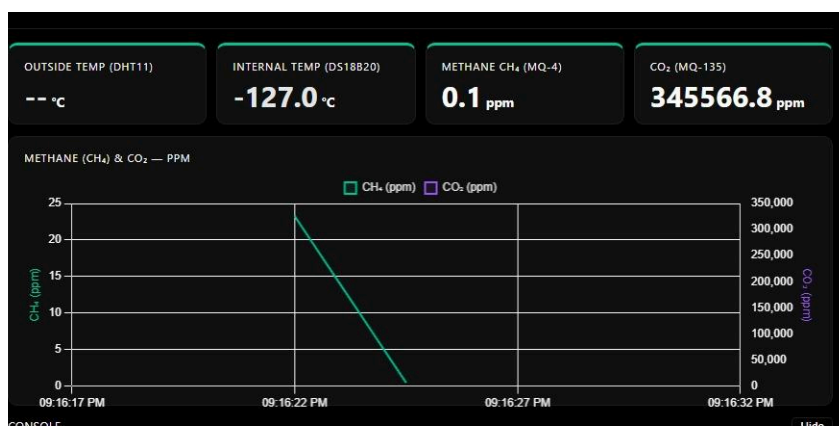
Figure: Digestomatic Electrical Architecture.

3.3 Testing & Problems

Firstly, the software was originally structured within a single main.ino file. However, due to code clutter and debugging difficulties, the project was refactored by splitting functionality into separate files.

Secondly, properly implementing the MQ-4 (CH₄) and MQ-135 (CO₂) sensors required several weeks due to the extensive calibration time needed for accurate readings, and even then, the results were not entirely seamless.

Thirdly, because each sensor operated on its own independent timing cycle, a custom timer was programmed to synchronize all measurements accurately. Sensor testing revealed that CO₂ and CH₄ (methane) concentration readings were highly inaccurate, causing the graphs to render improperly or fail entirely. As shown in the two images below, there are clear issues with both data accuracy and graph continuity.



Test run 19/7/2026
(The temperature sensors weren't activated and "-127 °C" is just the default value of the DS18B20 probe)

These problems were resolved by recalibrating the sensors, adjusting the calculation constants in the Arduino code, and updating the dashboard frontend to handle missing data points without crashing the charts.

3.4 Proof of Work

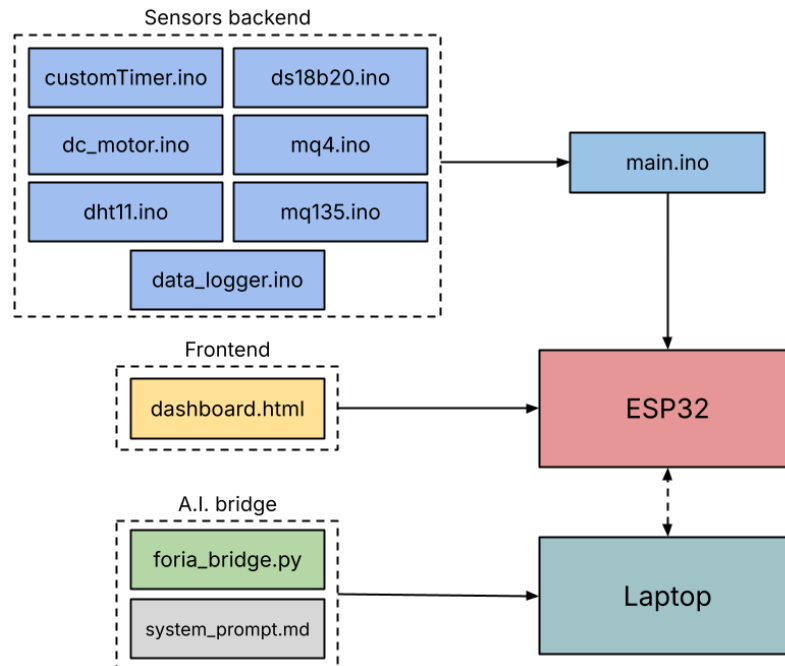


Figure: Software Architecture of Digestomatic.

Furthermore, a lot of Arduino/ESP32 libraries were necessary to facilitate the backend of this project:

- Adafruit Unified Sensor
- Async TCP
- DHT Sensor Library
- DallasTemperature
- ESP AsyncWebServer
- MQUnifiedsensor
- OneWire
- PubSubClient

And the following Python libraries are necessary:

- requests
- paho-mqtt
- fastapi
- pydantic
- uvicorn

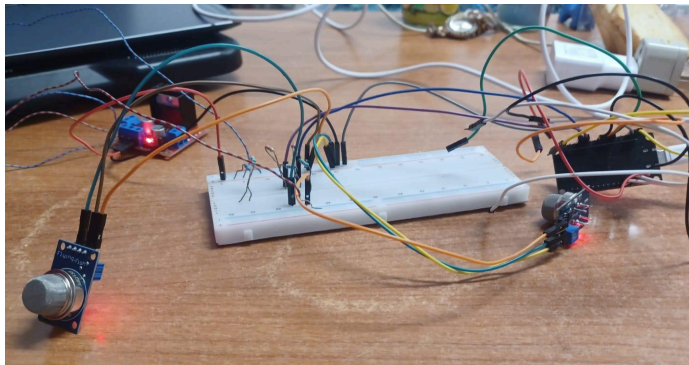


Figure: Test-run with gas sensors on breadboard. (22/7/2026)

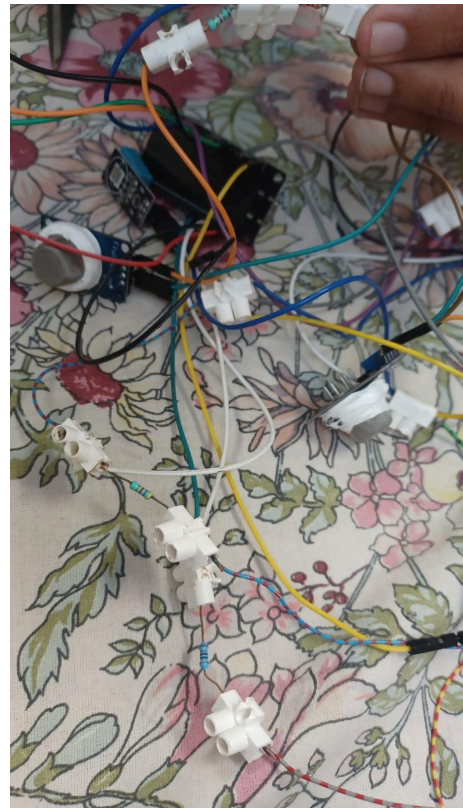


Figure: Application of screw terminal blocks on wiring. (1/8/2026)



Figure: Our group leader brainstorming on how to best fit the wiring into the biodigester's enclosure. (1/8/2026)

3.5 Files & Code

Below is a snippet of the main file 'main.ino' containing the setup, loop and other important functions. To see all files and code pertaining to Digestomatic, please see the [GitHub repo](#).

main.ino	Main file snippet containing main setup and main loop logic for stirrer, comms, sensors and data logging.
<pre>#include <WiFi.h> #include <ESPAsyncWebServer.h> #include <PubSubClient.h> // + rest of files (dht11.ino, dc_motor.ino, etc.) // WiFi, MQTT Config & Topics const char *WIFI_SSID = "...", *WIFI_PASSWORD = "...", *MQTT_SERVER = "10.163.92.202"; // MQTT topic definitions (TOPIC_TEMP_OUT, TOPIC_METHANE, TOPIC_COMMANDS, etc.) WiFiClient espClient; PubSubClient mqttClient(espClient); AsyncWebServer server(80); CustomTimer sensorTimer(5000); void onMqttMessage(char* topic, byte* payload, unsigned int length) { } void reconnectMQTT() { /* ... connect to broker & subscribe to TOPIC_COMMANDS */ } void setup() { Serial.begin(115200); // ... initialize sensors, motor, SPIFFS, and WiFi connection server.on("/", HTTP_GET, [](AsyncWebServerRequest *r) { /* ... serve dashboard html */ }); server.on("/download-log", HTTP_GET, [](AsyncWebServerRequest *r) { /* ... serve sensor_log.csv */ }); server.begin(); mqttClient.setServer(MQTT_SERVER, 1883).setCallback(onMqttMessage); } void loop() { if (!mqttClient.connected()) reconnectMQTT(); mqttClient.loop(); stirrer.update(); if (sensorTimer.isReady()) { // read sensors & publish values (MQTT) // add data to .csv file (data logging function) } }</pre>	

3.6 References & Credits

- [1] R. Santos and S. Santos, "ESP32 DS18B20 Temperature Sensor with Arduino IDE (Single, Multiple, Web Server)," *Random Nerd Tutorials*. [Online]. Available: <https://randomnerdtutorials.com/esp32-ds18b20-temperature-arduino-ide/>. [Accessed: June 13th, 2026].
- [2] R. Santos and S. Santos, "ESP32 DHT11/DHT22 Temperature and Humidity Sensor with Arduino IDE," *Random Nerd Tutorials*. [Online]. Available: <https://randomnerdtutorials.com/esp32-dht11-dht22-temperature-humidity-sensor-arduino-ide/>. [Accessed: June 13th, 2026].
- [3] R. Santos and S. Santos, "ESP32 Control DC Motor Speed and Direction with L298N Driver," *Random Nerd Tutorials*. [Online]. Available: <https://randomnerdtutorials.com/esp32-dc-motor-l298n-motor-driver-control-speed-direction/>. [Accessed: June 14th, 2026].
- [4] EZLO, "MQ4 Methane Gas Sensor," *Raspberry Pi EZLO*, Tutorials & Kits. [Online]. Available: <https://pi.ezlo.com/tutorials-and-kits/mq4-methane-gas-sensor/>. [Accessed: June 20, 2026].
- [5] IoT Web Planet, "ESP32 with an MQ-135 Gas Sensor," *IoT Web Planet*. [Online]. Available: <https://iotwebplanet.com/esp32-with-an-mq-135-gas-sensor/>. [Accessed: June 20, 2026].

4 Front-end & Platform

4.1 Objectives

To develop a user-friendly interface that communicates with both the AI and the biodigester's stirrer mechanism. The interface will feature dedicated widgets to display:

- *Live Temperature*: Internal and external temperature of the biodigester in °C;
- *Gas Levels*: Real-time CO₂ and CH₄ (methane) concentrations in ppm;
- *Historical Data*: Interactive graphs displaying CO₂ and CH₄ trends over time;
- *Project Information*: A link to the project's GitHub repository and an "About Us" section detailing the development team.

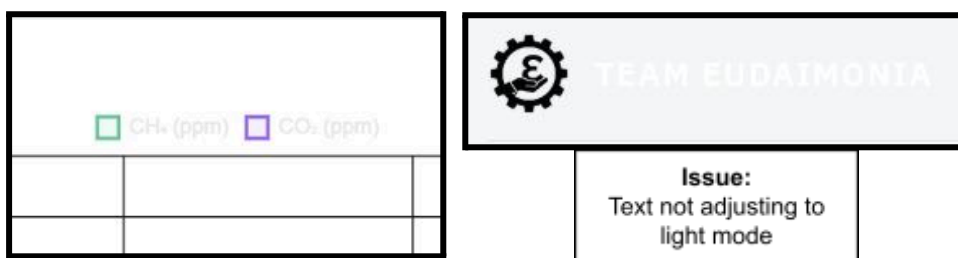
4.2 Steps Taken

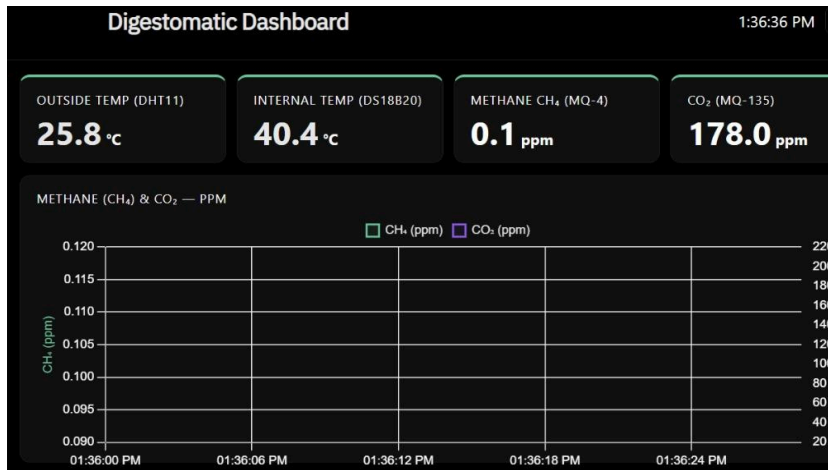
Using Arduino IDE v1.8.19, our team configured a SPIFFS file system on the ESP32 to host the web dashboard's HTML assets. We first built the UI skeleton, incorporating placeholder widgets for real-time graphs, sensor data, and the AI assistant. To improve visual appeal and user experience, we implemented several UI enhancements: integrated a light/dark mode toggle, placed our team logo and name in the top-left header, and applied a cohesive green theme to buttons and widget borders to reflect the natural concept of digestion.

Instead of making the dashboard's Mosquitto MQTT broker strictly Wi-Fi dependent (which was the original plan), we opted for local communication with Wi-Fi as a secondary option. Local comms means the system can work off-grid and ensures the user retains full, reliable control over their hardware. Our AI assistant, Foria, is powered by the Llama 3.2 model hosted locally using Ollama. Running the AI on local hardware keeps the entire system self-contained, highly responsive and avoids paid & limited API calls.

4.3 Testing & Problems

During development of the dashboard, we encountered many aesthetic issues regarding the layout and structure of the page. For example, certain icons or text in dark mode wouldn't be visible in light mode. But that was easily fixed. The main problem with the dashboard was that the methane and CO₂ chart didn't update properly, which meant the lines of the chart weren't showing up.

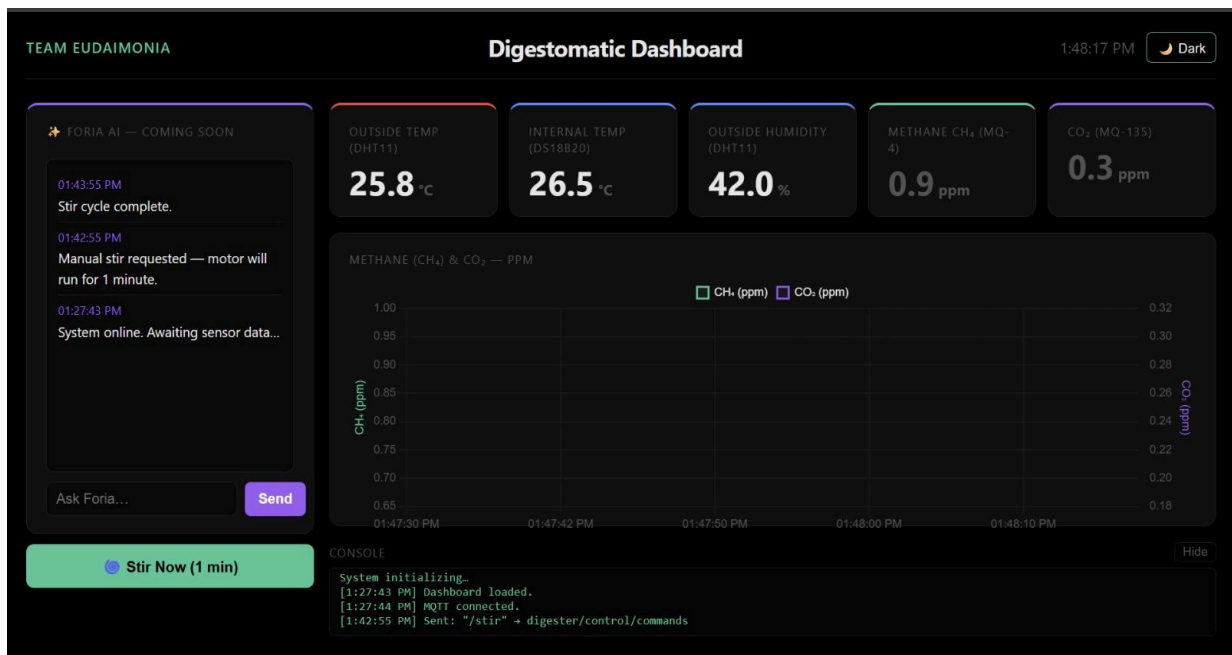




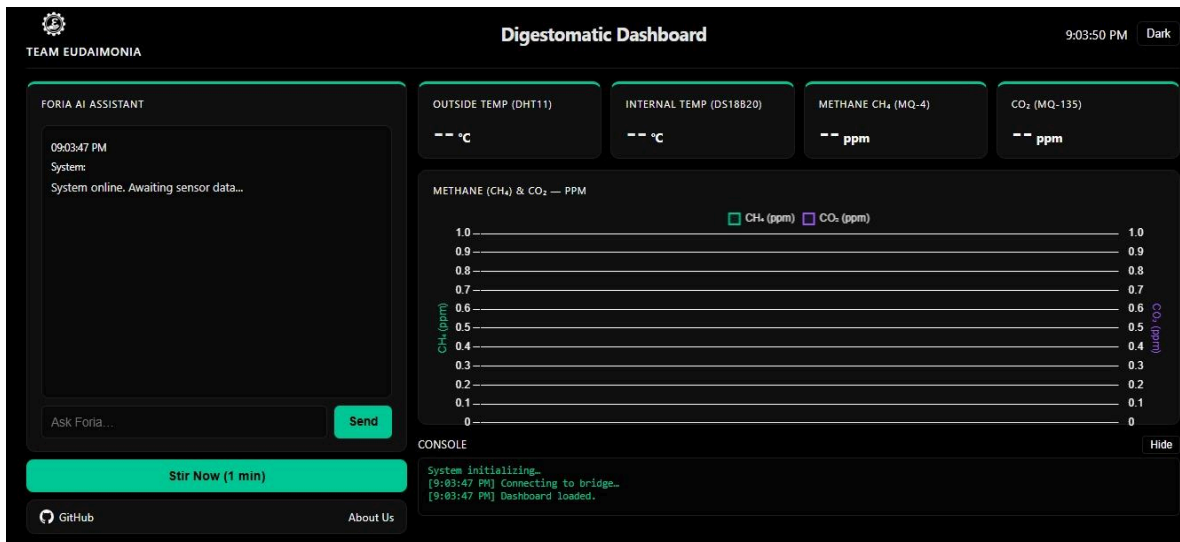
Test Run 16/7/2026
Issue: Chart lines not visible

4.4 Proof of Work

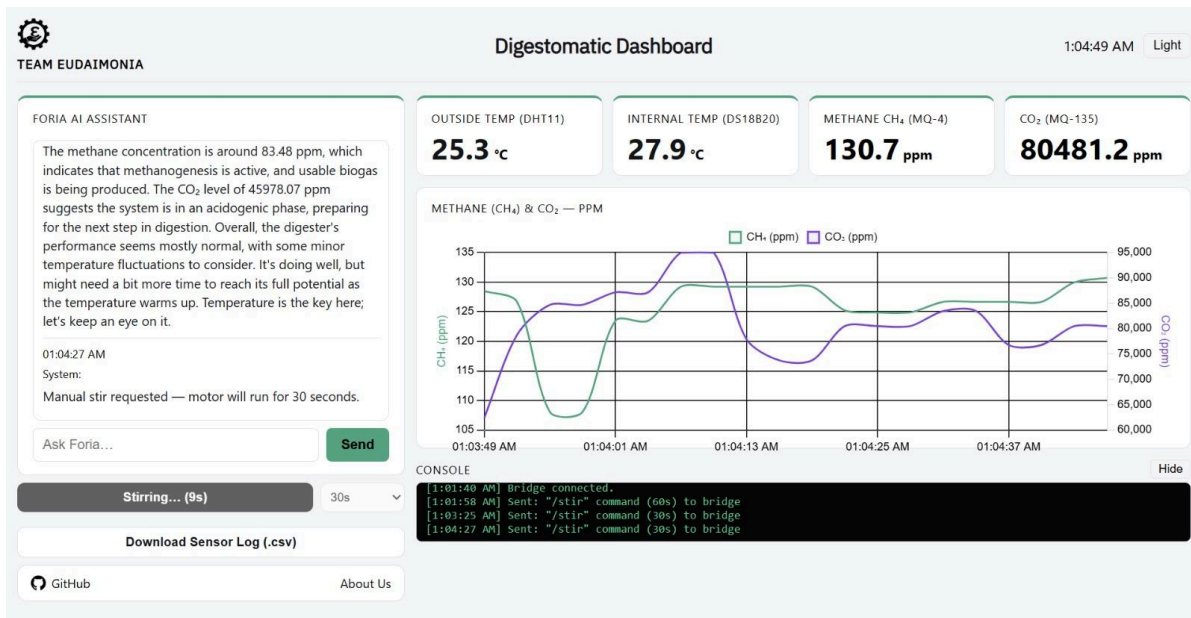
The following images are our Dashboard's designs throughout its development:



Dashboard design v0.67
Date of screenshot: 3/7/2026



Dashboard design v0.80
Date of screenshot: 14/7/2026



Dashboard design v1.0
Date of screenshot: 2/8/2026

4.5 Files & Code

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Digestomatic Dashboard</title>
7      <script src="chart.js"></script>
8      <style>
9          @font-face {
10             font-family: 'Inter';
11             src: url('Inter_18pt-Regular.ttf') format('truetype');
12             font-weight: 100 600;
13         }
14         @font-face {
15             font-family: 'IBM Plex Sans';
16             src: url('IBMPlexSans-Bold.ttf') format('truetype');
17             font-weight: 400 900;
18         }
19         :root {
20             --bg: #000000;
21             --surface: #0a0a0a;
22             --card: #111111;
23             --border: #1f1f1f;
24             --text: #e8e8e8;
25             --muted: #ffffff;
26             --accent-r: #ff3b3b;
27             --accent-g: #00c896;
28             --accent-b: #3b8bff;
29             --accent-y: #f5a623;
30             --accent-p: #9b59f5;
31         }
32
33         body.light-mode {
```

Figure: Dashboard's html code on GitHub
[dashboard_incl_foria.html](#)

4.6 References & Credits

[1] A. Menhennett and P. F. Navarro, *A Guide to HTML5 and CSS3*, Zenva, 2014. [Online]. Available: <https://drive.google.com/file/d/0BwwL5RPpTcRNRWNTVWZPRFhGSGJ6Q2xrbkE1VE5KUDBib2JN/view>. [Accessed: April 17, 2026].

[2] Ollama, "Ollama," [Online]. Available: <https://ollama.com/>. [Accessed: July 5, 2026].

[3] Eclipse Foundation, "Eclipse Mosquitto," *Mosquitto*. [Online]. Available: <http://mosquitto.org/>. [Accessed: July 5, 2026].

5 Business & Marketing

5.1 Objectives

The objective of the business aspect of Digestomatic is to commercially launch and position it within the Surinamese market, offering an alternative, more sustainable solution.

5.2 Market & Segment Analysis

One of the main goals of Digestomatic is to enhance the user experience of biodigesters. Designed as a scalable, plug-and-play appliance, Digestomatic serves as a complementary source of renewable energy. By processing daily organic waste into usable biogas on-site, it improves localized waste management and reduces reliance on non-renewable gas without having customers replace their existing household setup.

In Suriname, all gas comes from Ogane N.V, a subsidiary of Energiebedrijven Suriname N.V (EBS). The liquid petroleum gas (LPG) comes from non-renewable sources. It is imported and stored by Ogane in tanks. The gas is then transferred to cylinders and distributed by Ogane within the country. The problems are apparent: dependency and a single point of failure in case import, storage or distribution is halted. Digestomatic offers a sustainable solution to these problems.

Key Selling Points

Digestomatic's main competitor in Suriname is SlimGas N.V., currently the only other local biodigester provider. SlimGas offers passive, non-instrumented digesters, run as a one-person operation. Digestomatic differentiates itself as follows:

- Smart monitoring and control: real-time methane, CO₂, and temperature sensing (MQ-4, MQ-135, DHT11, DS18B20) paired with an automated stirring system, compared to a purely passive digester with no feedback or control loop.
- Adaptive stirring algorithm: an optional AI-driven mode analyzes a moving average of methane sensor readings to detect rate-of-change trends and adjusts stirring frequency accordingly. This targets crust formation on the slurry surface, which the literature indicates traps gas and reduces mixture homogeneity, and therefore supports more efficient digestion.
- Modular, serviceable design: electronics can be accessed or removed for maintenance without halting production. A second parallel line further reduces downtime risk for commercial customers.
- Scalability across one product line: the same core sensor package extends from household to industrial tanks, with only the motor, gearbox, and cabling changing by size, allowing one design to serve multiple customer tiers.

- Optional yield-boosting additives, such as biochar or ferric oxide, as a potential premium add-on.
- Stronger information and support infrastructure: a company website, a larger team, and structured customer education and consultation, compared to a one-man operation.
- Retrofit option: the sensor and control package can be sold on its own to upgrade an existing passive digester tank, lowering the entry barrier for customers who already have basic infrastructure.

Pricing Model

Digestomatic uses a modular, scale-based pricing model rather than a single fixed price, since the tank and its supporting hardware are built to the customer's needs. Two product lines exist within this model: full digester systems, and a tech-only retrofit package for customers who already own a passive tank.

Component tiers

- Electronics, standard tier: based on the prototype's sensor cost of roughly USD 20 to 30, translating to approximately SRD 1,000 retail with a safety margin for import cost fluctuation.
- Electronics, premium/business tier: uses more accurate, more resistant, and more durable sensors, priced above the standard tier. Exact pricing is not yet available and depends on supplier and component selection; flagged as an open item for the team to quote.
- Motor, gearbox, and stirrer assembly (propeller and shaft) sold as one integrated unit, scaling with tank size. Available in both electronics tiers, though the premium tier is not expected to be necessary for most household customers.
- Housing/enclosure for the electronics: estimated at up to SRD 1,000.

Reference: local tank costs

To ground the household pricing estimate, current retail prices for comparable durotanks in Suriname (Lucky Store, Rhino brand) were used as a cost reference for the tank component alone:

Tank size	Volume	Retail price (SRD)
100 gallon	378 L	3,450
200 gallon	756 L	6,500
450 gallon	1703 L	11,995
1,000 gallon	3785 L	33,995

Household system pricing

For an approximately 100-gallon household system (sized for a 4-person household as a starting assumption), a rough bottom-up build-up looks like this:

Component	Est. cost (SRD)	Confidence
100-gallon tank	~3,450	Firm (market price)
Standard electronics package	~1,000	Firm (prototype-based)
Housing/enclosure	up to 1,000	Estimate
Motor, gearbox, stirrer assembly	1500	Depends on size
Installation labor	1000	Effort/time based

Tier / segment	Indicative price (SRD)
Household, standard electronics (~100 gal)	15,000
Household, premium electronics	15,000 +
Larger household / small business (~200–450 gal)	Scales above 15,000 with tank size
Commercial / industrial	From ~100,000, scaling further with buyer requirements, no fixed ceiling
Tech-only retrofit package	Scales with existing tank size, same standard/premium tiers

Software and maintenance

- Basic software: timer-based stirring, one-time payment, includes free updates.
- Premium software (AI-driven): subscription-based, since the cloud-based adaptive stirring model carries ongoing hosting and compute cost. Basic customers can upgrade by paying the difference.
- Maintenance: subscription-based regular checks as the default, with on-demand paid repair/check-ups available outside the subscription for customers who opt out of it.

Target Audience (Customer Segments)

Primary segments

- Eco-conscious and cost-conscious households, particularly those with less convenient access to LPG cylinder points (for example, farming households in areas such as Nickerie, further from distribution).
- Restaurants and small businesses generate consistent food or organic waste, motivated primarily by direct gas-cost savings.
- Small-scale farmers and livestock farmers, who generate feedstock on-site and benefit from energy self-sufficiency.

Secondary / longer-term segments

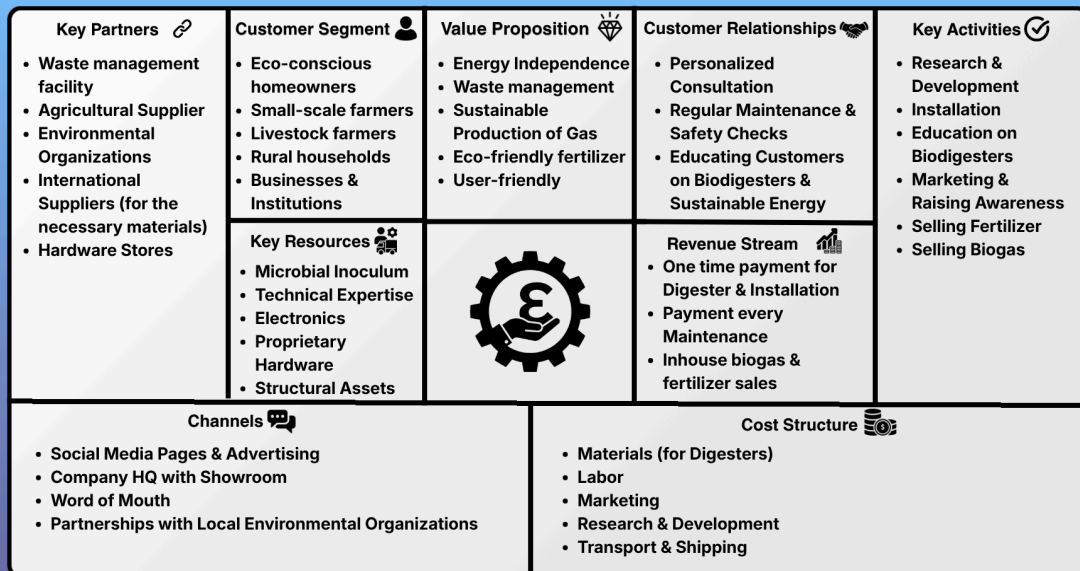
- Larger commercial and industrial operations (the 100,000 SRD+ tier).
- Institutional or B2B partnerships, including EBS/Ogane and septic-service companies such as Bron Bere (see Revenue Streams).
- Underserved neighborhoods or lower-income households currently priced out of consistent gas access.
- Inland villages: realistic mainly as a government development project or an NGO-driven humanitarian/modernization initiative, not as a near-term commercial segment, given access and infrastructure barriers.

Revenue Streams

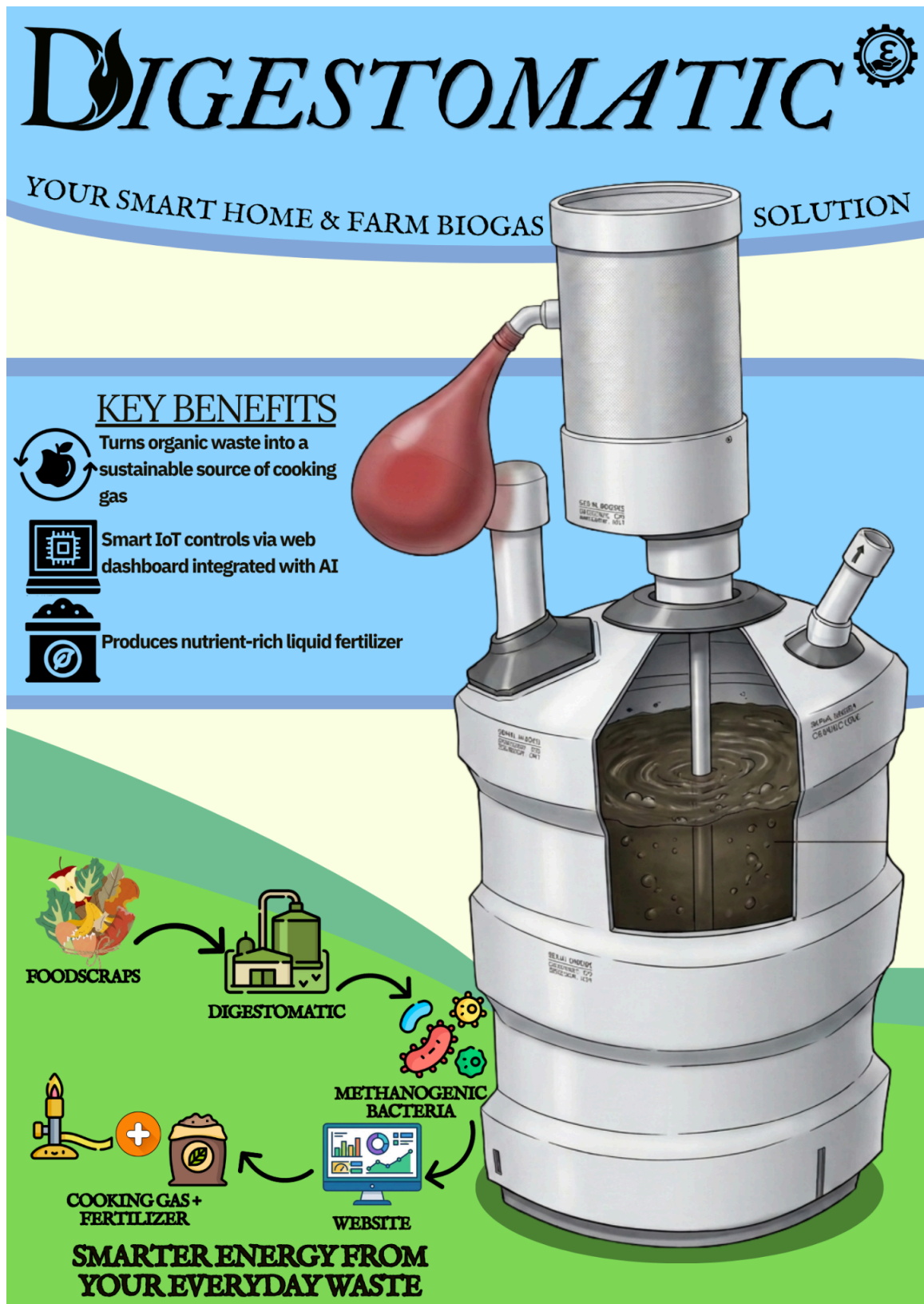
- One-time hardware sales: full digester units or tech-only retrofit packages, priced by scale from household to commercial.
- Software subscriptions: the premium AI-driven stirring and monitoring tier.
- Maintenance revenue: subscription-based regular checks, plus on-demand repair fees for customers outside the subscription.
- Feedstock partnerships: companies such as Bron Bere gain an easier sludge disposal route, either as a paid disposal service Digestomatic offers them, or as a fee they pay scaled to how burdensome their current disposal process is. This converts their waste stream into Digestomatic's feedstock input.
- Institutional/B2B with EBS or Ogane, two possible structures:
 - Joint venture: shared production and revenue, easier access to EBS's existing infrastructure and distribution, but closer integration and higher operational stress for Digestomatic.
 - Direct B2B sale: EBS purchases systems outright to produce their own biogas from pumped sludge. Simpler and lower risk, but lower long-term upside than a joint venture.

- Distribution: local sales handled directly; international or larger-scale growth would likely require distributors, though this remains an open question for future work rather than a settled plan.

5.3 Business Canvas



5.4 Project Poster



5.5 References & Credits

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6 Blog

Date	30 Apr 2026	Team Members	Everyone ▾
A session in which teams discuss the project among themselves, with the goal of creating a parts list and an AI integration plan. We already had the list of requirements. However, there was still no definitive plan regarding how to integrate the AI. We would like the AI to “predict” what will happen within the system based on the measurement data (temperature and gas concentration). It must also control a motor so that the mixture is processed optimally by the bacteria. In addition, it can send status messages that are displayed on the webpage regarding the conditions inside the biogas digester. This improves user-friendliness. Whether all of this is possible remains to be determined. For now, we as a team are focusing on the idea pitching. Margaret and Wongsonadi will work on the slides. Margaret is working on the Business Model Canvas (see this link).			

Date	1 May 2026	Team Members	Everyone ▾
A GitHub repository has been created containing all important files related to the project. By ‘important files’, we mean all documents, 3D models, drawings, schematics, lists, programs, and photos and/or videos created during the design and construction of the smart biodigester. The name of the project: Digestomatic. It was devised by group member Sprangers, and we voted to see if everyone agreed. For the time being, we are continuing to work on the idea pitching presentation. The date has been set for Saturday, May 9.			

Date	6 May 2026	Team Members	Everyone ▾
- Worked on a framework (first 3 slides) for the idea pitching slides on May 4, which allowed Salarbux and Wongsonadi to continue working on the rest. - Created a block diagram of Digestomatic (available on our GitHub). - Developed a timeline together with the rest of the team. - Salarbux completed the 3D prototype model. - Margaret created the Business Model Canvas. - Student Sprangers is currently learning HTML and CSS (for the dashboard).			

Date	9 May 2026	Team Members	Everyone ▾
On behalf of the group, Chris presented our project idea (Digestomatic: a smart biodigester). After the pitch (approximately 3 minutes), the judges asked their questions and made comments.			

The points raised and questions asked concerned feasibility within the 8 prototype building weeks, the implementation of a stirring mechanism, and the AI aspect. All valid comments. It takes 2-4 weeks for the bacterial population within the digester to be sufficient for anaerobic digestion to take place. Only at that point does methane production begin to increase. To accelerate that process, we will therefore use organic waste from septic tanks (poop, basically) to halve the 'incubation period'.

Regarding the stirring mechanism, it is essential to prevent a crust from forming in the top layer of the digester substrate (which is often a problem in passive biodigesters). In such a case, methane production stagnates. By stirring the mixture based on, for example, decreasing methane concentration, Digestomatic can keep methane production consistent.

Adding A.I. to a system that predicts what will happen can be very difficult. Given that this requires a lot of accurate measurement data regarding the conditions within our digester and that there are only 8 weeks available, we have devised another option. By simplifying an LLM (similar to ChatGPT, Gemini, or Claude) into a simpler and smaller model, specialized for Digestomatic, and integrating it with the system, the user experience can be improved.

Now that the project has been officially approved, we are awaiting further instructions from the organization and Mr. Boomsma.

Date	16 May 2026	Team Members	Everyone ▾
On May 14, all groups had already received certain sensors, motors, and other electronics from Mr. Boomsma. Team Eudaimonia only received the temperature probe (DS18B20), because the rest of the sensors were out of stock and had to be ordered. The groups are not required to do everything independently; supervisors are available, as well as equipment in the IoT lab at Etnalaan (3D printer, soldering irons, etc.). Mr. Boomsma told us about Mr. Samuel Kuik, better known as Slimmie. His company, SlimGas N.V., builds (passive) biodigesters. Today we visited his HQ, located somewhere on Nieuwe Zorgweg in Pontbuiten. Mr. Boomsma was there as well. Group leader Salarbux made contact first. After some back-and-forth communication, Slimmie indicated that he is willing to assist with building the project. Mr. Slimmie gave clear instructions regarding every step of Digestomatic (see photo below). Basically, the tank needs to be built differently, because the methane bacteria are very sensitive to light, humidity, and oxygen, among other things. We are currently updating the list of supplies and purchasing everything (that is not provided by IT Core).			
			

Date	22 May 2026	Team Members	Everyone ▾
The WB team members (Margaret, Salarbux, and Sprangers) started building the digester tank last Wednesday, May 20. They took a 20-liter water tank (without a handle) and sawed off the top section (where a pump is sometimes attached so that water can be pumped out into a bottle or something similar). Furthermore, they looked for the correct size drill cup so that holes could be made for the gas inlet and outlet. Additionally, Chris and Elijah worked on the implementation of the temperature sensors, namely the DHT11 and DS18B20. The DHT11 is placed outside the tank and the DS18B20 inside. The measurement data from both sensors were verified against those of a Pico HumidiProbe. Information regarding the accuracy of both was also reviewed in the datasheets. Based on all of this, it was concluded that we can proceed with the sensors for our project.			

Date	27 May 2026	Team Members	Jozua, Riaasat & Vaughn ▾
On May 23 and 24, work was done on splitting the programs for the temperature sensors (DHT11 and DS18B20). By using a file for each sensor, everything becomes clearer for the team, the judges, and anyone who wants to view our code. This utilized the 'branching' functionality of GitHub. The web dashboard works, but the layout, text, and colors still need to be formatted a bit more neatly. The WB part of the team has continued building the digester tank, but it is not yet fully completed. The 'effluent' outlet and the holes for sensors and the slurry agitator are still missing. We have not yet received an update from Mr. Boomsma regarding the rest of our supplies, including the MQ-4, MQ-135, and pH sensors. So, for the time being, we are simply continuing with what we can do.			
			

Date	22 Jun 2026	Team Members	Chris, Jozua, Riaasat & Vaughn ▾
We have finally received the remaining supplies from Mr. Boomsma, namely a DC motor with high torque and low RPM, an L298N motor driver, an MQ-4 methane sensor, and an MQ-135 gas sensor. We can get the pH probe, but on Mr. Boomsma's advice, we have chosen not to implement it. The probe cannot remain in the substrate constantly, and it would also cost an extra thirty dollars. We would also have to implement additional control besides the stirrer to fully utilize the pH readings. For now, the parameters are: methane concentration, CO2 concentration, internal temperature, and external temperature. Chris & Elijah will proceed with the IoT part so that the tank can be completed. After some discussion, we decided to build the stirring mechanism and monitoring system together in a kind of small circular lid. We call it the control unit.			

Date	3 Jul 2026	Team Members	Chris & Elijah ▾
This week, time was spent interfacing the MQ-4 and MQ-135 sensors with the ESP32. There was a burn-in period of approximately 36 hours during which both sensors were connected to the 5 V output of the L298N motor driver. Afterwards, on Thursday, July 2, both sensors were kept in the open air for about thirty minutes so that they could be calibrated. Fortunately, this could simply be done in open air, and no methane or anything similar was required for the MQ-4. The datasheets contain the information needed for self-calibration. Using a 24 V source supplying 1 A, as the entire system has a maximum current consumption of approximately 550 mA. However, everything shut down because 24 V at the input is too high for the L298N, which actually operates with an input voltage of 12 V. The operation of the temperature, gas detection, and CO2 detection implementations was checked. Furthermore, there is a button on the dashboard for manual stirring. A few 'final touches' are needed before everything can be handed over to the WB students to integrate with the tank, such as improving the web dashboard. The A.I. also still needs to be implemented.			

Date	9 Jul 2026	Team Members	Chris ▾
All necessary software installations (Ollama, llama 3.2, and Python libraries) were performed in advance, between the Idea Pitching and before all sensors arrived. This allowed for rapid work on adding AI functionality to the web dashboard. The name Foria was chosen, derived from Aeriforia, the Greek word for sustainability. Foria can answer questions about the biodigester in English or Dutch. I am currently trying to fine-tune everything, but there is a major problem that still needs to be solved: powering everything. The motor L298N motor driver has a 78M05 voltage regulator on it that outputs 5 V/500 mA. That is more than enough for all sensors. However, the plan is to power the ESP32 as well. The 500 mA supplied is not enough. Moreover, the L298N driver nominally operates at 12 V, because the 78M05 works optimally at 12 V. However, my tests show it can go up to about 18 V. If that limit is exceeded, the voltage regulator switches off. Why can't 12 V be used? Then the voltage is too low for the motor; it won't rotate. Or it will rotate very slowly. It's regarding current, max. 1 A not enough. All this means that I now have to start looking for an 18 V/1 A DC source, an LM2596 buck converter, a long USB cable, and/or all three. Regarding the tank, the Webers (WB students) are finished with the second tank.			

Date	16 Jul 2026	Team Members	Chris, Elijah & Vaughn ▾
Chris received a PC Power Supply Unit (PSU) (Thanks Mr. Kartopawiro), which was used to provide everything with power. The A.I. and IoT part is now working optimally; only some refinement is needed. The MQ-135 measurement data is not yet very accurate. Team member V. Sprangers is working on correcting it. The WB team members are continuing with the design and construction of the first tank. Team member E. Wongsonadi has started with the Hackomation documentation.			

Date	28 Jul 2026	Team Members	Everyone ▾
- Received new DS18B20, because the old one broke due to an error with grounding when testing with the PC PSU. - Made everything more compact, by ditching the breadboard and soldering the voltage dividers, etc. Also utilized stripped cables and electrical tape. - The digester tank Team further worked on the stirrer, and control unit. - Everything is operational. Just needs to be put together for the final test before the finale. - Margaret started on the poster.			

Date	29 Jul 2026	Team Members	Everyone ▾
- Elijah verified the integrity of all cable connections. - Vaughn made a logo for Digestomatic. - Jozua, Riaasat and Vaughn finished the tank. - Chris connected the PSU to the IoT network. - Everything (excluding PSU) was placed into the control unit housing. - The first main test of Digestomatic is scheduled for Saturday 1 August.			

Date	1 Aug 2026	Team Members	Chris, Jozua & Riaasat ▾
- Before we could start on the test, we had to correct a large problem with the stirrer. The propeller needed to be fastened very tightly, because otherwise it gets loose due to mechanical oscillations. - That ended up taking many hours as we tried many different methods first. - Finally, we just used epoxy and tightened the screws extremely.			

Date	2 Aug 2026	Team Members	Everyone ▾
- Sealing everything and connecting everything took many hours, but after it was done, we started the first test of Digestomatic. - Everything worked on the first try! - Results from yesterday and today were shared with Vaughn for the documentation. - The entire system was tested a second time and this time the .csv data was downloaded.			

Date	5 Aug 2026	Team Members	Everyone ▾
- Jozua, Elijah, Vaughn and Riaasat worked and finalized the poster and it was sent to Mr. Boomsma on 4th of August. - There was an issue with the motor; it didn't rotate anymore. After some troubleshooting, we narrowed the problem down to the motor driver and adjusted the wiring and program to use the other inputs and outputs. - We spray painted the tank. - Everything is operational now. - Made a demo video and uploaded it to YouTube (see 1.2 Demo for the link). - Further work on docs and demo for the finale.			